USE CASE, DATA & TEST SPECIFICATION

Use cases, acceptance criteria and initial verification plan

AI-Powered Multimodal Assistive Application for People with Mild Cognitive Impairment (MCI)

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1. Use Case Catalogue

ID	Use Case	Actor	Precondition	Success Outcome
UC-01	Authenticate User	Patient/Caregiver	Valid account exists	Successful authentication and role-specific home screen.
UC-02	View Daily Schedule	Patient	Schedule exists	Today's activities displayed.
UC-03	Manage Schedule	Caregiver	Authorized patient selected	Schedule saved/updated.
UC-04	Manage Medication	Caregiver	Authorized patient selected	Medication and timing stored.
UC-05	Receive Medication Reminder	Patient	Active medication schedule exists	Reminder delivered at configured time.
UC-06	Find Misplaced Object	Patient	Object has a recent observation	Last detected location and time returned.
UC-07	Recognize Familiar Person	Patient/System	Person is registered; camera available	Identity and relationship cue displayed.
UC-08	Retrieve Conversation Memory	Patient/System	Relevant memory exists	Contextual memory retrieved and used in response.
UC-09	Ask Personal AI Assistant	Patient	Assistant service available	Answer returned from structured data and/or RAG.
UC-10	Set Safe Zone	Caregiver	Location data available	Geofence stored and activated.
UC-11	Handle Geofence Violation	System/Caregiver	Active geofence; location crosses boundary	Caregiver alert created and delivered.
UC-12	Play Well-being Session	Patient	Content available	Selected session plays.
UC-13	Manage Familiar Person	Caregiver	Authorized patient selected	Person profile/recognition data created or updated.
UC-14	Review Patient Activity	Caregiver	Authorized patient selected	Relevant activity history displayed.

2. Detailed Use Cases

UC-06 Find Misplaced Object

1. Patient selects or speaks the name of an object.
2. System searches recent object observations.
3. If a valid observation exists, the latest observation is selected.
4. System presents object name, last observed context/location and timestamp.
5. If no observation exists, system states that it has no recorded observation.

UC-07 Recognize Familiar Person

6. Camera captures a face.
7. System detects the face.
8. System creates a recognition representation.
9. Representation is compared with registered familiar people.
10. If confidence exceeds threshold, name and relationship are displayed.

11. Otherwise the person is treated as unknown.

UC-08 Retrieve Conversation Memory

12. Patient asks a memory-related question.
13. System converts voice to text when required.
14. System extracts available person/topic/time context.
15. System retrieves relevant vector memories.
16. Relevant memory is passed to the response generator.
17. System produces a contextual answer and avoids unsupported claims.

UC-11 Handle Geofence Violation

18. Caregiver has an active safe zone.
19. Patient device provides a supported location update.
20. System evaluates location against the safe zone.
21. If outside, a violation event is created.
22. Caregiver notification is generated.
23. Event is recorded in activity/alert history.

3. Acceptance Criteria

ID	Area	Acceptance Criterion
AC-01	Authentication	Correct credentials authenticate; incorrect credentials are rejected; role permissions are enforced.
AC-02	Schedule	Caregiver creates a schedule item and patient can see it.
AC-03	Medication	Medication schedule produces a reminder at the configured time.
AC-04	Object Finder	After a test object is detected and stored, querying the object returns its last observation.
AC-05	Face Recognition	A registered test person is recognized under supported lighting/distance conditions; unknown test face is not falsely mapped.
AC-06	RAG	A stored test conversation can be retrieved using a semantically related query.
AC-07	Person-Aware RAG	When a registered person's face is recognized, a related test conversation can be preferentially retrieved.
AC-08	Assistant	Assistant can answer a schedule/medication query using structured data without hallucinating missing fields.
AC-09	Geofence	Crossing a test boundary creates an alert and notification event.
AC-10	Privacy	Unauthorized caregiver/user cannot access another patient's protected information.

4. Initial Test Plan

ID	Test	Input/Condition	Expected Result	Type

TC-001	Login with valid credentials	Valid user account	Authenticated dashboard shown	Functional
TC-002	Login with invalid credentials	Wrong password	Error shown; no session created	Security/Functional
TC-003	Role access test	Patient account opens caregiver route	Access denied	Security
TC-004	Create medication	Caregiver enters valid medication	Record persists	Functional
TC-005	Medication reminder	Configured reminder time reached	Patient reminder generated	Integration
TC-006	Object detection	Supported object placed in camera view	Object detected and observation stored	AI/Functional
TC-007	Object retrieval	Previously stored object queried	Latest observation returned	Integration
TC-008	Known face	Registered person's face shown	Correct profile returned under test conditions	AI/Functional
TC-009	Unknown face	Unregistered face shown	Unknown/uncertain result	AI/Safety
TC-010	Conversation ingestion	Test conversation stored	Embedding/index record created	AI/Integration
TC-011	RAG query	Semantically related question	Relevant memory retrieved	AI
TC-012	Person-aware RAG	Known person + related query	Relevant person-associated memory ranked/retrieved	AI/Integration
TC-013	No-memory query	Question with no matching memory	System states lack of information	Safety/AI
TC-014	Geofence inside	Location within radius	No violation alert	Functional
TC-015	Geofence outside	Location beyond radius	Alert generated	Integration
TC-016	Unauthorized patient access	Caregiver without permission	Data access rejected	Security
TC-017	AI service failure	Disable LLM service	Core schedule/reminder functions remain available	Reliability

5. Data Validation Rules

- Required patient/caregiver identifiers shall not be empty.
- Medication name and schedule shall be present before an active medication reminder is created.
- Geofence radius shall be a valid positive value.
- Face profile shall be linked to a registered person.
- Object observation confidence shall be stored when provided by the model.
- Conversation memory records shall identify their source and timestamp.
- Sensitive data access shall be checked against the authenticated user and patient relationship.

6. Traceability Matrix — Initial

Requirement(s)	Area	Design Module	Verification
FR-003/FR-004	Schedule	MOD-03	TC-004/functional schedule tests
FR-005/FR-006	Medication/Reminders	MOD-04	TC-005
FR-009/FR-010/FR-011	Object Intelligence	MOD-05	TC-006/TC-007
FR-012/FR-014	Face Intelligence	MOD-06	TC-008/TC-009

FR-016/FR-017/FR-018/FR-019	RAG	MOD-07/MOD-08	TC-010/TC-011/TC-012/TC-013
FR-020/FR-021/FR-022	Safety	MOD-09/MOD-10	TC-014/TC-015
FR-001/FR-002	Security	MOD-01	TC-001/TC-002/TC-003/TC-016
NFR-005	Reliability	Cross-module	TC-017

7. Demonstration Scenarios

24. Scenario A — Lost object: Patient asks for keys; system retrieves the latest observation and speaks/displays the result.
25. Scenario B — Familiar person: Patient points camera at a registered person; system identifies name/relationship.
26. Scenario C — Contextual memory: Recognized person is associated with a query; system retrieves the relevant previous conversation.
27. Scenario D — Routine: Patient sees today's schedule and receives a medication reminder.
28. Scenario E — Safety: Test patient location leaves the safe zone; caregiver receives an alert.
29. Scenario F — Caregiver management: Caregiver updates medication/schedule and sees the updated status.

8. Project Completion Checklist

- Requirements baseline approved.
- Database schema reviewed.
- API specification finalized.
- Patient UI implemented.
- Caregiver UI implemented.
- Authentication/RBAC tested.
- Schedule/reminder/medication workflow tested.
- Object pipeline tested.
- Face pipeline tested.
- RAG ingestion/retrieval tested.
- Person-aware RAG integration tested.
- Geofencing tested with simulated/controlled locations.
- Notification flow tested.
- Privacy/security controls tested.
- Final documentation and demonstration scenarios completed.